Coordinated Wildfire Inspection with a Battery-Constrained Drone Fleet

Group 7 — Research Dossier
COMP6008 Reinforcement Learning
For independent study and report preparation — NOT FOR SUBMISSION

Policy notice: The unit specification prohibits generative AI in submitted work. This AI-assisted source-checked research dossier must not be submitted verbatim; all members must independently write the assessed report, verify citations and attribute their work. COMP6008 | Group 7 | Working research dossier for independent report preparation

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Reviewed against July 29, 2026 guidelines and repository master at b9f6b7438afafcb25455f707ecda7b004d6fc457.

ABSTRACT—Evidence synopsis. A single centrally controlled fleet of three drones performs recurring inspections of simulated wildfire hazards while managing shared charging, battery depletion, evolving fire, and inspection deadlines. DQN, REINFORCE, and synchronous A2C are implemented and evaluated in a fully observable, finite-horizon Markov decision process. A frozen 40×40 comparison comprises nine independent trained policies and 260 test missions. Greedy achieves 0.22077 mean weighted freshness, versus DQN 0.08438, REINFORCE 0.06901, and A2C 0.08435. Masked random yields 0.08379. Zero losses for masked learners therefore evidence strong rule-based feasibility control, not a demonstrated learned surveillance advantage. The notes below identify verified methods, limitations, and literature-grounded follow-ups; they are for independent student synthesis, not direct submission.

INDEX TERMS—reinforcement learning; wildfire monitoring; Markov decision processes; drone coordination; DQN; policy gradient; actor–critic; policy evaluation.

I. Project description and critical review

A. Problem framing and research aim

The approved project asks how three battery-limited drones can repeatedly inspect known, changing wildfire hazards over a finite mission. The controller must trade inspection freshness against transit energy, overdue locations, scarce charging capacity and collision restrictions. This is not a hidden-fire discovery experiment: fire, fuel, terrain, priority, wind and drone condition are fully observed in the simulation [1], [12]. Novelty is a coordinated sequential decision problem, not the claim that deep RL necessarily wins against engineered patrol rules. Related peer-reviewed drone-fire studies justify the application domain but do not calibrate the parameters of this simulator [9], [10].

Why nontrivial: an action that increases coverage now can strand a drone later; high-priority sites change with burning cells and fire front; three simultaneous moves can interfere; and all drones share one charger. A one-step urgency rule may nonetheless be formidable. The suitable scientific question is whether learned policies yield measurable sustained inspection improvement under an unchanged task and comparable held-out scenarios.

B. Fully observable finite-horizon MDP

Decision process: M=(S,A,P,r,γ,H). The observation is a Gymnasium dictionary: five full-grid maps—burning, normalized fuel, terrain susceptibility, static priority, and inspection age—of shape (5,n,n); three records [row,column,battery,lost] of shape (3,4); and 12 global entries for wind (five one-hot categories), charger ownership (four), base coordinates (two), and remaining-time proportion (one). In the main study n=40 and H=300. Each variable is normalized to [0,1]. The neural encoder additionally constructs four binary location maps from these observed coordinates, yielding nine spatial channels and 24 nonspatial inputs. It does not supply hidden future fire information [12], [13].

One central action controls all three drones. Each receives one of stay/north/south/west/east; the joint discrete space therefore has 5^3=125 actions. Inspection happens automatically after movement. At the shared base, a staying drone can charge, but only one charger owner is serviced per step. A physical transition resolves simultaneous moves and blocked collisions, applies energy and charge updates, increments and resets inspection ages, computes reward, then evolves fire and wind. Full fleet loss or the horizon ends an episode. Transition randomness is seeded; parameters are illustrative rather than physically calibrated [12].

The 40×40 environment uses battery capacity 160, movement cost 1, hover cost 0.25, recharge rate 8, radius 2, inspection deadline 30, initial three fires, base near the grid centre, fire-spread probability 0.04 and wind-change probability 0.06. Three higher-priority zones are initialized

per scenario. Terrain influences fire, not obstacles to flight. The single controller cannot be equated with three independent learning agents.

C. Objective, reward and measurable outcomes

The simulator computes location priority w_i = p_i(1 + 2 b_i + f_i), where p_i is static site priority, b_i marks burning cells and f_i marks a fuel-bearing fire-front neighbour. At each step, weighted freshness is F_t = $\sum$_i w_i max(0,1-age_i/deadline) / $\sum$_i w_i. The overdue fraction is the weighted proportion with age_i > deadline. These are quality metrics on a [0,1] scale, not the percentage of area covered [12].

The environment reward is the sum of weighted inspection gain, a negative overdue penalty, normalized energy cost, blocked collision attempts and newly lost drones, plus a remaining-horizon overdue charge on total fleet failure. Frozen weights are +1.0 gain, -0.1 overdue fraction, -0.02 × energy/160, -0.2 per blocked command and -5 per new drone loss. This reward is shaped in the general sense of combining terms, but the main 40×40 experiment sets additional coordination shaping to zero. Distinguish this existing multi-term objective from a proposed potential-based shaping method [8].

Report distinct metrics: full-horizon weighted freshness, overdue fraction, episode return, energy, loss count, collision attempts, fleet-failure rate, charge completions and safe-return events. A safe return event is not an assertion that all drones ended at base. On an early total fleet failure the evaluator pads unserved timesteps with zero freshness and fully overdue locations rather than rewarding a shortened run [1], [12].

D. Selection and implementation of three RL methods

DQN (value-based): a convolutional encoder maps the full observation to 125 joint Q-values; replay samples transitions; a target network supplies bootstrapped TD estimates; epsilon-greedy data collection explores admissible candidate actions. The final corrected masked DQN uses a feasible next-action TD maximum and standard DQN targets, not Double DQN [2], [13].

REINFORCE (Monte Carlo policy gradient): a categorical policy over feasible joint actions collects full episodes, then weights log-probabilities by discounted reward-to-go minus an optional learned Monte Carlo baseline. It contains no TD bootstrap, despite sharing the neural encoder with the other methods [3], [13].

A2C (synchronous actor–critic): one network returns categorical policy logits and a state-value estimate; 32-step rollouts use bootstrapped discounted targets to update policy and critic. Its entropy penalty regularizes action selection; it is synchronous single-environment A2C, not the asynchronous A3C algorithm described by Mnih et al. [4], [13].

Algorithm–code mapping to independently validate: environment reset/step in src/wild_drone/environment.py; action_mask and masked TD target in masked_learning.py; neural encoder, return and TD utilities in learning.py; candidate and Greedy/sweep policies in baselines.py; experiment runner, audit and replay scripts in scripts/. The consolidated Wildfire_Drone_Fleet.ipynb already provides implementation-oriented pseudocode; students must verify and attribute each assessed code/notebook section.

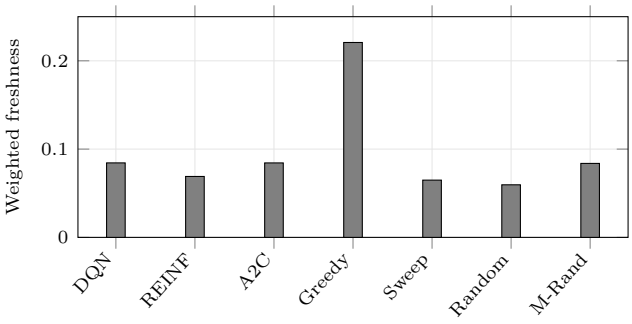


Fig. 1. Main-study freshness means; illustrative graph redrawn from published study statistics [1].

E. Experiments and training decisions

Historical development used 20×20 worlds to diagnose poor unmasked survival, repeated commands, instability and incomplete charger coordination. Changes included spatial drone/base channels, REINFORCE's Monte Carlo baseline, feasible-action rules, lower DQN learning rate, Double DQN screening and diagnostic supervised probes. These experiments must not be merged with the final 40×40 numerical estimates or described as successful RL tuning without evidence [1], [14], [15].

The frozen main comparison used independently initialized training seeds 0, 1, 2 for each of DQN, REINFORCE and A2C (nine fits). Each completed 300 episodes on training worlds 0–299 (90,000 environment interactions per fit, 810,000 total). All final checkpoints were evaluated on unseen worlds 22000–22019: 9×20=180 learned-policy missions plus 4×20=80 baselines, totalling 260. No 40×40 best-checkpoint selection, post-result tuning, supervised warm start or additional coordination shaping was used [1], [11].

Shared settings include γ=0.99, Adam, gradient clipping 0.5 and a common spatial input. DQN: learning rate 1×10^{-4}, minibatch 32, replay capacity 2,000, target refresh every 100 update steps and epsilon 1→0.05 over 40,000 steps. REINFORCE: learning rate 1×10^{-3} and entropy weight 0, with a learned Monte Carlo baseline. A2C: learning rate 1×10^{-4}, 32-step rollout and entropy weight 0.05. These are fixed study choices, not proof of optimum settings [11].

F. Frozen 40×40 outcomes

The Greedy policy has approximately 2.62× DQN's mean freshness and 3.20× REINFORCE's. DQN improves

TABLE I
Audited main-study evaluation means across 260 missions.

Policy	Fresh. ↑	Overdue ↓	Return ↑	Loss ↓
DQN	0.08438	0.86869	-24.79	0
REINFORCE	0.06901	0.89402	-25.81	0
A2C	0.08435	0.86949	-24.83	0
Greedy	0.22077	0.65563	-16.10	0
Sweep	0.06486	0.90121	-26.07	0
Random	0.05956	0.92098	-42.87	3
Masked random	0.08379	0.88601	-25.46	0

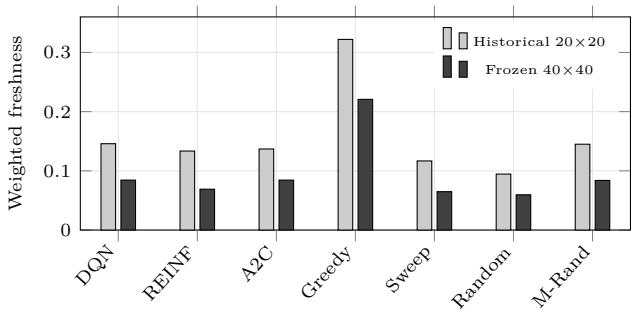


Fig. 2. The 20×20 and 40×40 values come from different code/configuration regimes; do not infer an isolated scaling effect [14].

over masked random by only 0.000589 freshness (0.059 percentage points); A2C by 0.000553; REINFORCE is below it by 0.014780. All learned methods miss the prespecified all-seed contribution and Greedy capability gates. An intact fleet at the horizon did not imply good monitoring. Unmasked random loses all three drones in each of its 20 test missions, whereas all nine masked fitted policies and masked random have zero average losses [1].

Statistical caution: each learned controller's average first aggregates 20 worlds within each fitted seed, then averages three seed means. Freshness seed-level sample SD is DQN 0.00254, REINFORCE 0.00730 and A2C 0.01163. Baseline variability is across 20 worlds, a different unit. Three fitted replicates are insufficient for strong significance or convergence claims. DQN's +0.000589 average difference is descriptive, not evidence of a meaningful positive effect [1].

G. Critical review: separating task completion from RL success

The implementation and audit objectives were substantially met: an explicit MDP, independently implemented three-method comparison, notebook-linked pseudocode, multiple historical pilots, fresh-world evaluation, and a completed 40×40 training protocol. Independent replay checked all 75,911 final evaluation steps, selected actions, masks, state hashes and rewards, with zero neural-score drift. This supports reproducibility of the recorded experiment, not validation of real wildfire physics [1], [11].

The central scientific finding is negative. DQN and A2C lie close to a policy that uniformly samples rule-approved actions; REINFORCE performs worse. Feasible-action filtering blocks unsafe battery/collision commands and is a dominant explanation for survival. Because Greedy uses its native candidate logic and supervised probes have separate objectives, the comparison is not a perfectly identical algorithmic control. The absence of validated learning superiority must be stated plainly [1], [14].

Potential mechanisms supported by historical diagnostics include a large 125-way joint action head, difficult long-horizon inspection/charging credit assignment, weak learned ranking under distribution shift, repeated argmax actions and (in earlier unmasked 20×20 runs) very large Q predictions relative to bounded return. The 40×40 audited experiment does not identify one unique causal mechanism. Earlier pilot changes were not isolated one-factor causal estimates, and additional compute alone does not resolve why a Greedy controller wins [14], [15].

Construct validity is limited: true fire and wind are observed; the grid ignores smoke occlusion, uncertain sensors, no-fly zones and calibrated physical units; fire evolves through synthetic stochastic cellular rules. External validity to actual UAV emergency operations has not been established. Also, the 20×20 and 40×40 comparison changes horizons, energy settings, sensor radius, inspection deadline and source revision, so the observed numerical difference is not a controlled scalability result [12], [14].

II. Potential improvements and future expansions

Guideline distinction: an improvement should address a measured weakness in the present system, while future work should expand the scientific task beyond that weakness. Parameter-only changes do not count as improvements. At least three of each are needed for a three-person group, with at least six qualifying scholarly sources and around 40% of total report space for Part II. The items below are hypotheses and research routes, not implemented achievements [16].

A. Improvements tied directly to current results

1) Improvement 1 — Structured multi-drone credit assignment: Evidence of limitation: the existing central network predicts one number for each of 125 simultaneous

move combinations. At 40×40, its fitted policies rarely improve weighted freshness beyond masked random despite conserving all drones. A plausible issue is that a shared scalar return does not readily reveal which drone should inspect, return or yield the charger; this diagnosis is not proven solely by the aggregate scores.

Proposed experiment: replace the monolithic joint policy head with a structured cooperative decision model that estimates per-drone utilities conditioned on shared state and recombines them through an explicit coordination operator (e.g., a QMIX-style mixing network). Alternatively evaluate a central critic with counterfactual per-agent advantage to separate one drone's contribution from teammates' actions (COMA). The current task remains centralised execution; these methods would be adapted accordingly rather than falsely described as already implemented [6], [7].

Why science supports it: Rashid et al. show factorised value functions can recover coordinated joint action selection under monotonic mixing, while Foerster et al. develop counterfactual baselines to improve cooperative multi-agent credit assignment. Those findings motivate a hypothesis, not an expected magnitude of improvement for this simulator. Compare equal environment steps, seeds and mask rules; report charger queues, duplicate coverage, regret to Greedy and held-out freshness. Test an ablation that preserves the old encoder and varies only action/critic structure. An admissible negative outcome would be unchanged freshness, indicating the main bottleneck lies elsewhere [6], [7].

2) Improvement 2 — Potential-based timeliness signal: Evidence of limitation: the objective rewards current inspection gain and penalises overdue sites, yet all RL methods fail to organise persistent patrols. A carefully designed learning signal could give early credit for positioning toward a high-priority overdue cluster without changing the task's intended optimal policy. However, earlier ad hoc shaping probes in the repository produced negative findings; their outcomes must be retained rather than rewritten as progress [14].

Proposed method: define a bounded state potential $\Phi(s)$ based on priority-weighted inspection urgency, feasible expected travel distance to serviceable regions, and charger waiting condition. Add shaping $F(s,a,s')=\gamma\Phi(s')-\Phi(s)$ only in training; hold physical outcomes and the original test reward fixed. Ng et al. establish policy invariance under this potential-difference construction subject to relevant MDP assumptions [8]. Explicitly handle finite-horizon termination with a consistent terminal potential so an agent cannot earn artificial extra credit by ending early.

Test design: freeze the new potential before opening test worlds, compare no-shaping versus potential shaping under matched initialization, interactions and feasible-action rules, and audit original reward separately from shaped learning return. Plot time-to-first-return, average weighted age, overdue fraction and learning curves. A positive result requires improvement in held-out original-objective freshness across seeds, not merely increased shaped training reward. The method is a new reward-structure intervention, not learning-rate tuning [8].

3) Improvement 3 — Robust learning on learner-visited states: Evidence of limitation: historical supervised inspection probes scored highly on their training rankings but failed after rollout, consistent with a mismatch between demonstrated states and self-generated patrol trajectories. A later loss-only ranking change reduced full-mission freshness from 0.19200 to 0.13559, showing that increasing a training matching score does not necessarily improve end-to-end behaviour [14], [17].

Proposed method: study an explicit dataset-and-policy iteration process that retains dangerous return/charger states, uses controlled learner-state collection, and applies conservative offline value estimation to avoid unsupported action values. Conservative Q-Learning (CQL) motivates penalising overly optimistic values for unsupported actions, while prioritised experience replay motivates sampling rare consequential transitions more often [18], [5]. This is not a claim that a supervised classifier or a CQL implementation already solves the project; CQL's offline assumptions must be checked before adapting it to online mission data.

Evaluation: preregister a matched dataset size, control with uniform replay, and separate (i) held-out decision agreement from (ii) autonomous full-horizon outcomes. Publish the proportion of training packets covering return-to-base, collision-avoidance, charging-queue and high-overdue events, along with weighted freshness, losses and drift under fresh scenario seeds. Repeat across three or more fits. Evidence of progress is robust closed-loop monitoring, not near-perfect labels on static sampled states [5], [18].

B. Future directions extending the problem scope

1) Extension 1 — Partial observability and belief-based sensing: Different research question: the current controller sees complete fire maps even before any visit. A realistic extension would hide true ignition, introduce uncertain camera detections and smoke-induced observations, and require the fleet to maintain a belief map of fire and uncertainty. This changes the problem from the present fully observed MDP to a partially observable decision process, testing both where to inspect and what has actually been learned from sensing. Hauskrecht surveys the approximation challenges of POMDP value functions; Julian and Kochenderfer investigate imperfect sensing in distributed wildfire surveillance [9], [10].

Proposed experimental design: add a sensor likelihood $p(o|s)$, detection/false-alarm rates, delayed information updates and a held-out belief-calibration metric; retain the physical task's existing motion rules for the first study. Compare age-based patrol, information-gain planning and learned belief policies under identical sensing budgets.

Report true-state coverage, belief error, time-to-detection and inspection freshness. The claim would concern information gathering, not a post hoc fix to the existing 40×40 RL deficit.

2) Extension 2 — Decentralised fleets with lossy communication: Different research question: today one central controller has instantaneous access to all drones and the single charger. In a wider multi-UAV mission, each aircraft may have its own local sensors, limited radio contact and intermittent communication. Introduce a Dec-POMDP or cooperative Markov-game formulation with per-drone observations and delayed messages. Compare local-only, periodic shared summaries and learned communication protocols; quantify bytes transmitted, stale information and safety under outages [6], [19].

Scientific basis: coordinated value factorisation (QMIX) and differentiable/reinforcement-based communication (DIAL/RIAL) show principled ways to research shared-team decisions under imperfect information [6], [19]. The proposed benchmark must include fair baselines, communication delay distributions and collision/battery compliance—not merely more agents on a larger map. This extension concerns the autonomy/coordination architecture rather than repairing current one-step ranking errors.

3) Extension 3 — Calibrated, heterogeneous wildfire operations: Different research question: assess whether a policy class can generalise to physically meaningful mission conditions. Introduce geographically referenced terrain, calibrated spread and wind fields, distinct sensor footprints, heterogeneous batteries and additional chargers. Consider uncertainty and multi-objective priorities such as timely tracking of an advancing front versus visiting fixed infrastructure. Viseras et al. evaluate deep Q-learning for cooperative fire-front monitoring, while Julian and Kochenderfer address wildfire surveillance with autonomous aircraft, providing application-specific research precedents [9], [10].

Study design: reserve entire weather regimes or map regions as held-out domains, and report risk-constrained timeliness, energy, completion, robustness to sensor failures and simulated real-time decision latency. Compare physically calibrated baselines alongside RL rather than scaling the present synthetic grid and calling it field-ready. Domain randomisation or a digital twin may be investigated, but no operational readiness should be claimed without external measurements and safety review. The existing project remains an educational simulator, not a validated wildfire-response system [9], [10].

C. Prioritisation and scientific decision rules

Experimental discipline for any follow-up: archive the source revision, model variant, random seeds, world allocation, simulation budget, baseline implementation, and primary metric before fitting. Tune only on declared training/validation scenarios; evaluate once on a fresh final set; preserve negative trials and report both failures and costs. For this already completed assignment, none of these proposed experiments needs to be misrepresented as finished.

III. Individual contributions: group completion required

The guideline requires an identified author for every project-code and report section, agreed individual contribution percentages, and transparent attribution. Repository commit authorship alone does not prove agreed assessed contributions, and this document does not invent names or shares. The group must complete the following collaboratively before any independently written submission [16].

Required check: annotate each assessed notebook section with its actual contributor(s); mark the beginning of each independently written report section with its author; confirm the percentages sum to 100%; obtain agreement from all members. If contributions are equal, the group may state this explicitly only if it is true. Confirm unit coordinator approval and Blackboard updates independently.

Reference library: independently verify sources

References

[1] Wild Drone Appeared, "Frozen 40×40 RL main study," docs/experiments/rl-main-2026-10-11/README.md and summary.csv, master revision b9f6b743, 2026. https://github.com/SayuruRehan/wild-drone-appeared

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[7] J. N. Foerster et al., "Counterfactual multi-agent policy gradients," in Proc. AAAI Conf. Artificial Intelligence, vol. 32, pp. 2974–2982, 2018, doi: 10.1609/AAAI.V32I1.11794.

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[11] Wild Drone Appeared, "Frozen main-grid RL comparison," docs/scale-rl-protocol.md; "rl-main-frozen.json," configs/, 2026. Repository as in [1].

[12] Wild Drone Appeared, "Implementation design" and environment.py, docs/design.md and src/wild_drone/environment.py, 2026. Repository as in [1].

TABLE II
Improvement and extension planning summary.

Proposal	Current evidence / change	Success check
I1	Near-masked-random; factorised / counterfactual fleet credit	Repeated unseen freshness, coordination events
I2	Poor patrol timeliness; policy-invariant potential reward	Higher original reward/freshness, no new violations
I3	Weak closed-loop generalisation; conservative data iteration	Autonomous 300-step mission gains, not just label fit
E1	Hidden-fire sensing; POMDP with belief map	Belief calibration and true detection latency
E2	No comm constraints; local cooperative policies	Performance vs message delay/bandwidth
E3	Synthetic physics; heterogeneous calibrated missions	Robustness on held-out weather/maps

TABLE III
Contribution assignment placeholders (group confirmation required).

Member	Notebook / code sections	Report sections	Agreed share
Student A: _____	_____	_____	_____ %
Student B: _____	_____	_____	_____ %
Student C: _____	_____	_____	_____ %
Total			100 %

[13] Wild Drone Appeared, "Learning and masked-learning source," src/wild_drone/learning.py and masked_learning.py, 2026. Repository as in [1].

[14] Wild Drone Appeared, "20×20 and 40×40 comparison" and "Completed T4 pilot," docs/20x20-40x40-comparison.md and docs/colab1000-results.md, 2026. Repository as in [1].

[15] Wild Drone Appeared, "Learning pilot" and "Study integrity fixes," docs/learning-pilot.md and docs/verification-2026-10-10.md, 2026. Repository as in [1].

[16] Curtin University, "COMP6008 Reinforcement Learning: Group Project, Presentation, and Report Guidelines," Jul. 29, 2026, pp. 1–11. User-supplied specification; check for Blackboard revisions.

[17] Wild Drone Appeared, "Frozen loss-only 40×40 comparison," docs/experiments/scale-loss-2026-10-11/README.md, 2026. Repository as in [1].

[18] A. Kumar, A. Zhou, G. Tucker, and S. Levine, "Conservative Q-Learning for offline reinforcement learning," in Advances in Neural Information Processing Systems, vol. 33, 2020. https://proceedings.neurips.cc/paper/2020/hash/0d2b2061826a5df3221116a5085a6052-Abstract.html

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Appendix A
Assignment alignment and final verification

This appendix is an internal checklist, not assessment writing. The 29 July 2026 brief says generative AI is strictly prohibited in any part of submitted work, including report, code documentation and presentation. Students must independently read the primary evidence, write and verify their final text, and demonstrate understanding during the interview. The report deadline stated in the brief is 30 October 2026 at 23:59, with the code and presentation deadline 23 October; Blackboard may supersede these dates [16].

Appendix B
Code and evidence pointers

Wildfire_Drone_Fleet.ipynb — Single consolidated notebook; mathematical/pseudocode mapping and frozen results.

src/wild_drone/environment.py — Observation, action encoding, state updates, reward and episode termination.

src/wild_drone/learning.py — Neural encoder, discounting, TD/Monte Carlo utilities and unmasked learning.

src/wild_drone/masked_learning.py — Feasibility selection, masked DQN targets, training/evaluation.

src/wild_drone/baselines.py — Greedy, sweep, random and planner baselines.

configs/rl-main-frozen.json — Exact declared 40×40 settings, training seeds and final-world IDs.

docs/experiments/rl-main-2026-10-11/summary.csv — Full metrics including uncertainty units and decision speed.

docs/20x20-40x40-comparison.md — Historical differences, caveats and outstanding assessed deliverables.

docs/experiments/scale-loss-2026-10-11/README.md — Negative supervised loss probe; must not be described as a successful RL method.

docs/scale-rl-protocol.md — Prespecified 40×40 comparison, success gates and audit controls.

Appendix C
Questions for independent oral defence

1. Which fields make the environment fully observable, and why is the task not unknown-fire discovery?
2. How do 5^3 joint actions encode movement and how do collision/charge rules constrain their effects?
3. Can you derive the exact reward terms from the implementation and explain early-failure accounting?

TABLE IV
Assignment alignment and outstanding verification checks.

Guideline requirement	Verified evidence / action still required
Three independently implemented RL methods	DQN, REINFORCE, A2C in Python; all nine 40×40 fits complete
MDP state/action/transition/reward	environment.py, design.md and consolidated notebook; students verify mappings
Training and tuning analysis	20×20 development/Colab pilots + frozen 40×40 summary; distinguish protocols
Validation on unseen scenarios	20 new 40×40 worlds (22000–22019), all 260 missions replay audited
Critical analysis	Greedy strongly exceeds all learned controllers; safety mostly rule-associated
Improvement proposals	At least 3 weakness-linked structural methods; not mere parameter changes
Future expansions	At least 3 distinct scope extensions; minimum six academic works
Report presentation	Use official IEEE two-column A4 Word template; at most eight pages before refs/appendix
Contributions	Add real author names, section allocation and agreed percentages
Submission/assessment	Independently write, verify and prepare viva; observe no-GenAI rule

4. What distinguishes DQN's TD target, REINFORCE's full Monte Carlo return, and A2C's n-step bootstrap?
5. Why does zero drone loss not imply learning or successful charging coordination?
6. How is seed-level uncertainty calculated and why are 60 test missions not 60 independent fitted models?
7. Which 20×20 differences make the apparent scaling trend noncausal?
8. What evidence justifies each proposed improvement and what new evaluation would disprove it?
9. How would a partial-observation extension alter S, O, P and the policy input?
10. Which contribution percentages and notebook/report authorship have been agreed by the group?

Evidence and source metadata checked on 18 October 2026. Original code and experiment records remain in the private GitHub repository. This independent study packet is not an assessment-ready manuscript and does not guarantee similarity or integrity-policy compliance if copied into a submission.